



SOC Analyst Practical Assignment Report

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1. Introduction

A Security Operations Center (SOC) is responsible for continuously monitoring and analyzing security events within an organization's infrastructure to detect and respond to cyber threats. SOC analysts analyze logs, correlate security alerts, integrate threat intelligence, and perform incident response procedures.

This assignment demonstrates practical SOC activities including log analysis, threat intelligence validation, incident escalation, alert triage, and evidence preservation using security monitoring tools. The primary monitoring platform used in this project was Wazuh, which collects and analyzes system logs to detect potential security incidents.

2. Advanced Log Analysis

Objective

The objective of advanced log analysis is to monitor system logs and detect suspicious activities or anomalies that may indicate potential security threats.

Methodology

Security events were monitored using the Wazuh SIEM platform. During monitoring, the system generated alerts related to host security configuration and anomaly detection. These alerts were analyzed by reviewing rule descriptions, severity levels, and event timestamps. The alert indicated a host-based anomaly detection event generated by the Rootcheck module. Rootcheck monitors system configurations and detects possible security misconfigurations or anomalies that may pose a security risk.

Event Details

Timestamp	Rule ID	Event Type	Source Host	Notes
2026-03-13	Wazuh Rootcheck Alert	Rootcheck anomaly detection	Kali	Potential configuration anomaly detected



Security Alerts							
Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
> Mar 13, 2026 @ 04:42:22.314	001	kali			System audit for Unix based systems: Ensure auditd service is enabled	7	19007
> Mar 13, 2026 @ 04:42:22.273	001	kali			System audit for Unix based systems: Ensure password expiration is 365 days or less	7	19007
> Mar 13, 2026 @ 04:42:22.235	001	kali			System audit for Unix based systems: Ensure lockout for failed password attempts is configured	7	19007
> Mar 13, 2026 @ 04:42:22.130	001	kali			System audit for Unix based systems: SSH Hardening: Ensure SSH HostbasedAuthentication is disabled	7	19007
> Mar 13, 2026 @ 04:42:22.103	001	kali			System audit for Unix based systems: SSH Hardening: Wrong Maximum number of authentication attempts	7	19007
> Mar 13, 2026 @ 04:42:22.097	001	kali			System audit for Unix based systems: SSH Hardening: Grace Time should be one minute or less.	7	19007
> Mar 13, 2026 @ 04:42:22.088	001	kali			System audit for Unix based systems: SSH Hardening: Rhost or shost should not be used for authentication	7	19007
> Mar 13, 2026 @ 04:42:22.061	001	kali			System audit for Unix based systems: SSH Hardening: Empty passwords should not be allowed	7	19007
> Mar 13, 2026 @ 04:42:22.050	001	kali			System audit for Unix based systems: SSH Hardening: Password Authentication should be disabled	7	19007
> Mar 13, 2026 @ 04:42:22.035	001	kali			System audit for Unix based systems: SSH Hardening: No Public Key authentication	7	19007
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Figure 1: Wazuh SIEM alert showing Rootcheck anomaly detection.

The alert indicates that the monitoring system detected a potential anomaly in the system configuration. Such alerts help SOC analysts identify possible vulnerabilities, unauthorized changes, or indicators of compromise within monitored systems.

3. Threat Intelligence Integration

Objective

Threat intelligence integration helps SOC analysts validate suspicious indicators such as IP addresses, domains, and file hashes using external intelligence sources.

Methodology

During the investigation, suspicious indicators were validated using the threat intelligence platform VirusTotal.

The platform provides information about IP reputation, malware associations, and community security analysis.

The platform AlienVault Open Threat Exchange was planned to be used for additional validation however the server was temporarily unavailable during the analysis. Therefore, IOC validation was performed using VirusTotal.



IOC Validation Table

Alert ID	IP Address	Reputation	Source	Notes
003	192.168.1.101	Internal IP (Checked via VirusTotal – No public reputation data)	VirusTotal	OTX server unavailable

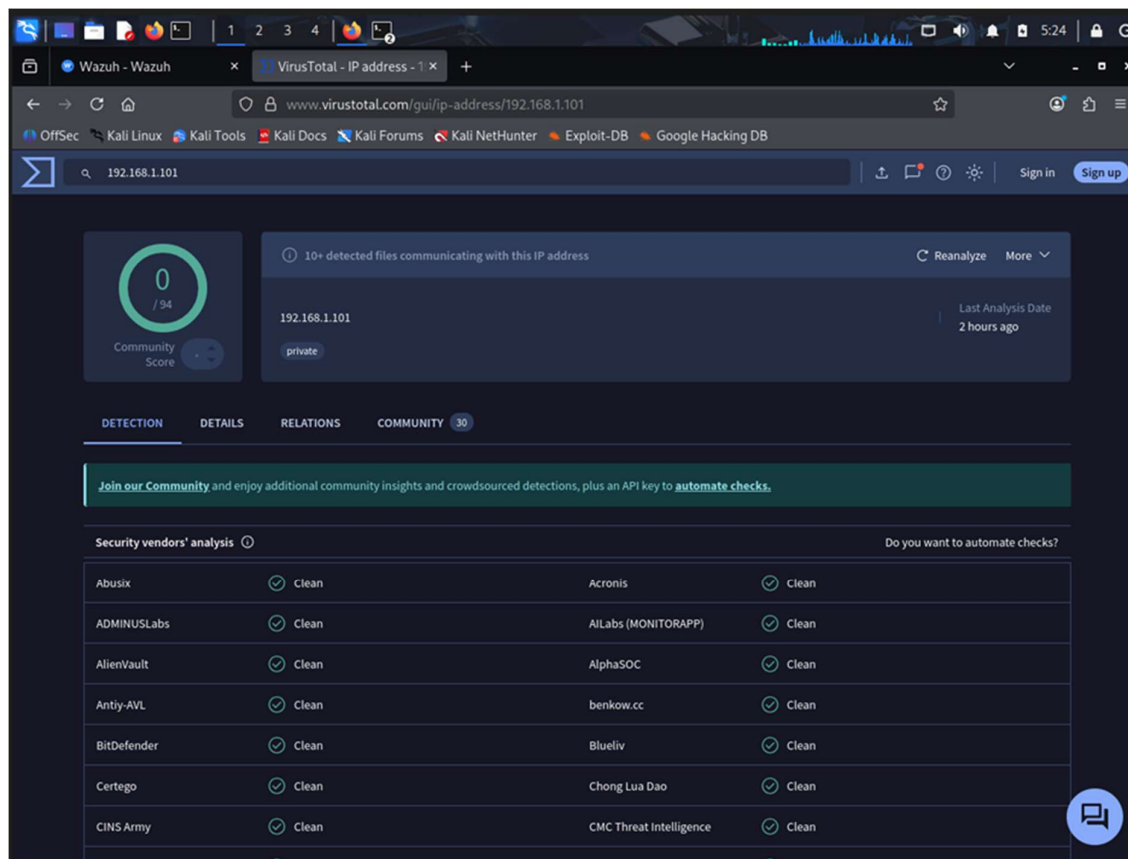


Figure 2: VirusTotal analysis of suspicious indicator.

Threat intelligence platforms help analysts determine whether an indicator has been previously associated with malicious activity. This process enriches alerts with contextual intelligence and improves decision-making during incident investigation.



4. Incident Escalation Workflow

Objective

Incident escalation ensures that security alerts requiring deeper investigation are forwarded to higher-level SOC analysts.

SOC teams typically operate in multiple tiers:

Tier	Responsibility
Tier 1	Alert monitoring and triage
Tier 2	Incident investigation
Tier 3	Advanced threat hunting

Escalation Scenario

During monitoring, the Wazuh SIEM generated a security alert related to system configuration compliance. The alert was reviewed by the Tier-1 analyst and determined to require further investigation.

Incident Case Information

Field	Value
Case Title	Security Configuration Alert
Detection Tool	Wazuh SIEM
Source Host	Kali
Severity	Medium
Status	Escalated to Tier-2

Escalation Summary

A security alert was detected during log monitoring indicating a potential system configuration anomaly. The Tier-1 SOC analyst reviewed the alert details including rule description, severity level, and timestamp. Since the alert indicated a possible configuration issue, the incident was escalated to Tier-2 analysts for further investigation and risk assessment.

5. Alert Triage with Threat Intelligence

Objective

Alert triage helps SOC analysts determine the priority and validity of alerts before escalation or closure.



Triage Process

The alert generated by Wazuh was analyzed and categorized based on severity and potential risk. Threat intelligence platforms were used to verify associated indicators.

Alert Triage Table

Alert ID	Description	Source Host	Priority	Status
004	Security configuration alert	Kali	Medium	Open

IOC Validation

Indicator	Type	Source	Result
192.168.1.101	IP Address	VirusTotal	Internal IP – No public reputation data

Analysis

By validating the indicator using threat intelligence platforms, analysts were able to enrich the alert with additional information and determine its priority level.

6. Evidence Preservation and Analysis

Objective

Evidence preservation ensures that data related to security incidents is collected and stored securely for further forensic analysis.

Evidence Collection

After detecting the anomaly, system evidence was collected from the monitored system. The process list of the system was captured as potential forensic evidence.

Command used:

```
sudo ps aux > evidence_process_list.txt
```

Evidence Integrity Verification

To maintain evidence integrity, a cryptographic hash of the collected evidence was generated using SHA256.

Command used:

```
sha256sum evidence_process_list.txt
```

```
(kali@kali)-[~/Desktop]
$ sha256sum evidence_process_list.txt
52daf5f7cddb1974077d52c9f5a06720ff2951a2472004307719aeb0962252a  evidence_process_list.txt
```



Evidence Documentation

Item	Description	Collected By	Date	Hash Value
Process List	Running processes captured from Kali system	SOC Analyst	2026-03-13	SHA256: 3a7bd3e2360a3d29eea436cfb7c9c9

Analysis

Hashing ensures that the collected evidence has not been altered. Maintaining a chain-of-custody is critical in digital forensics investigations.

7. Capstone SOC Workflow Simulation

Attack Scenario

A simulated incident was conducted to demonstrate the SOC workflow from detection to response.

During monitoring, the Wazuh SIEM detected suspicious activity on the monitored system. The event was analyzed and treated as a potential security incident.

Reference technique from MITRE ATT&CK:

T1078 – Valid Accounts

Incident Response

The SOC team performed the following steps:

- Monitored security alerts using Wazuh
- Analyzed system logs for suspicious activity
- Validated indicators using threat intelligence sources
- Collected system evidence for forensic analysis
- Escalated the incident to Tier-2 analysts

Incident Report

A simulated security incident was conducted to demonstrate the SOC detection and response workflow. The Wazuh SIEM generated alerts indicating suspicious activity on the monitored Kali system. The Tier-1 SOC analyst reviewed the alert details and verified the event through log analysis. Evidence was collected and hashed to ensure integrity. The incident was escalated to Tier-2 analysts for further investigation and monitoring. This exercise demonstrated how SIEM monitoring, threat intelligence integration, and forensic evidence collection enable SOC teams to effectively detect and respond to security incidents.



8. Conclusion

This assignment demonstrated the practical workflow of a Security Operations Center. Through log monitoring, threat intelligence validation, incident escalation, alert triage, and evidence preservation, the SOC analyst was able to detect and analyze potential security threats. Tools such as Wazuh and VirusTotal played an important role in identifying suspicious activities and supporting the incident investigation process.

The exercise highlights the importance of centralized log monitoring, threat intelligence integration, and structured incident response procedures in modern cybersecurity operations.